ENTHUSIAST ADVANCE COURSE

PHASE : MEC-I (B)

TARGET : PRE-MEDICAL 2024

Test Type : **MINOR** Test Pattern : **NEET (UG)**

TEST DATE : 14-08-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	3	4	1	2	4	3	4	3	3	3	4	4	2	2	2	2	3	3	4	2	4	1	1	2	4	3	3	3	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	2	3	1	1	1	3	2	2	4	4	1	1	2	3	1	2	2	2	3	2	1	2	1	2	2	4	3	3	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	1	3	1	3	4	4	2	1	3	4	2	3	1	4	2	4	3	1	2	2	2	4	2	1	1	3	1	4	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	4	3	2	4	2	4	4	4	1	2	2	2	1	3	1	1	1	3	4	2	2	1	1	4	3	2	3	1	1	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	2	4	1	3	2	3	4	1	2	2	1	1	3	2	4	3	2	2	2	1	2	2	4	4	3	3	4	4	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	3	1	3	2	2	1	1	2	1	2	2	3	3	3	4	3	2	2	1	2	3	3	3	1	4	2	3	3	3
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	3	3	1	2	4	4	4	3	4	3	1	1	1	3	1	1	4	4	3	1										

HINT – SHEET

SUBJECT : PHYSICS

SECTION-A

2. Ans (3)

$\alpha : \beta : \gamma = 1 : 2 : 3$

4. Ans (1)

$$\frac{\Delta T}{T} = \frac{1}{2} \propto \Delta\theta = \frac{1}{2} \times 5 \times 10^{-6} \times (30 - 10)$$
$$= 5 \times 10^{-5}$$

5. Ans (2)

$$I = I_1 + I_2 + 2\sqrt{I_1}\sqrt{I_2}\cos\phi$$
$$I_1 = I_2 = I'$$
$$I_{max} = 4I' = I_0$$
$$I' = \frac{I_0}{4} \quad \text{for } \Delta = \frac{\lambda}{3}, \phi = \frac{2\pi}{3}$$
$$I = \frac{I_0}{4} + \frac{I_0}{4} + 2\sqrt{\frac{I_0}{4}}\sqrt{\frac{I_0}{4}}\cos\frac{2\pi}{3}$$
$$I = \frac{I_0}{2} + 2 \times \frac{I_0}{4} \times \left(-\frac{1}{2}\right) = \frac{I_0}{2} - \frac{I_0}{4} = \frac{I_0}{4}$$

7. Ans (3)

n^{th} minima

$$x = \frac{n\lambda D}{a} \quad \{n = 1\}$$
$$x = \frac{\lambda D}{a} \Rightarrow a = \frac{\lambda D}{x} = \frac{5 \times 10^{-7} \times 2}{5 \times 10^{-3}}$$
$$a = 0.2 \text{ mm}$$

8. Ans (4)

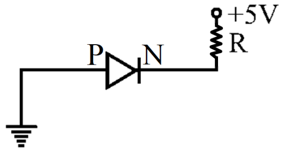
$$R.P = \frac{\alpha}{1.22\lambda}$$
$$R.P. \propto \frac{1}{\lambda}$$
$$\frac{(R.P.)_1}{(R.P.)_2} = \frac{\lambda_2}{\lambda_1} = \frac{5000}{4000} = \frac{5}{4}$$

9. Ans (3)

By the method of reflection all statements are correct.

17. Ans (2)

(a)

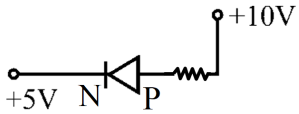


P ⟶ Low potential (L.P.) ⟶ 0V

N ⟶ High potential (H.P.) ⟶ 5V

So, D ⟶ R.B.

(b)

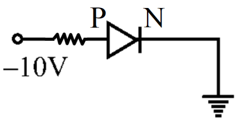


P ⟶ 10V (H.P.)

N ⟶ 5V (L.P.)

So, D ⟶ F.B.

(c)

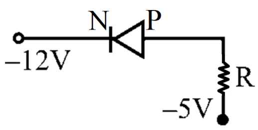


P ⟶ 10V (L.P.)

N ⟶ 0V (H.P.)

So, D ⟶ R.B.

(d)

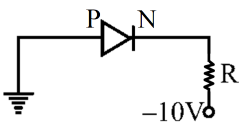


P ⟶ –5V (H.P.)

N ⟶ –12V (L.P.)

So, D ⟶ F.B.

(e)



P ⟶ 0V (H.P.)

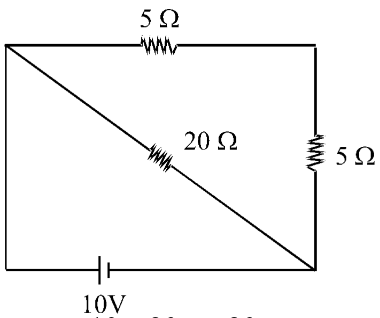
N ⟶ –10V (L.P.)

So, D ⟶ F.B.

So, (b), (d), (e) are forward biased.

18. Ans (3)

D_1 is in reverse bias, D_2, D_3 in forward bias.



$$R_{eq} = \frac{10 \times 20}{10 + 20} = \frac{20}{3}$$

$$i = \frac{10}{20/3} = 1.5A$$

19. Ans (3)

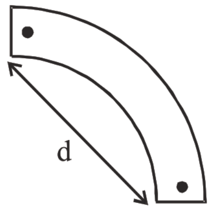
By using $r = \dfrac{mv\sin\theta}{qB}$

$$\Rightarrow r = \frac{1.67 \times 10^{-27} \times 2 \times 10^{6} \times \sin 30^\circ}{1.6 \times 10^{-19} \times 0.104} = 0.1\text{m}$$

and it's time period

$$= \frac{2 \times \pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.104} = 2\pi \times 10^{-7}\text{ sec.}$$

20. Ans (4)



$$\frac{2\pi r}{4} = L \quad \Rightarrow \quad r = \frac{2L}{\pi}$$

$$d = \sqrt{2}\,r = \frac{2\sqrt{2}\,L}{\pi}$$

$$M_{new} = \frac{(M.d)}{L}$$

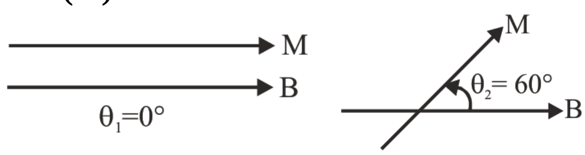
$$= M\left\{\frac{2\sqrt{2}}{\pi}\right\} = \frac{2\sqrt{2}\,M}{\pi}$$

21. Ans (2)

$$B_a = \frac{2kM}{d^3} = B; \quad B_e = \frac{kM}{d^3}$$

$$\Rightarrow B_e = \frac{k(8M)}{(2d)^3} = \frac{kM}{d^3} = \frac{B_a}{2} = \frac{B}{2}$$

22. Ans (4)



$$W = MB(\cos\theta_1 - \cos\theta_2) = MB(\cos 0^\circ - \cos 60^\circ)$$

$$= MB\left(1 - \frac{1}{2}\right) = \frac{MB}{2} \Rightarrow MB = 2W$$

$$\Rightarrow \quad MB = 2\sqrt{3}\text{ N-m}$$

$$\tau = MB\sin 60^\circ = (2\sqrt{3})\left(\frac{\sqrt{3}}{2}\right)\text{ J} = 3\text{J}$$

23. Ans (1)

$$\frac{mv^2}{r} = qvB \Rightarrow B = \frac{mv}{rq}$$

$$B = \frac{9.1 \times 10^{-31} \times 10^{6}}{0.5 \times 1.6 \times 10^{-19}} = 1.13 \times 10^{-5}\text{T}$$

24. Ans (1)

$M = \dfrac{qL}{2m}$

$L = \dfrac{MR^2\omega}{2}$

$M = \dfrac{1}{4} q\,\omega R^2$

25. Ans (2)

$r = \sqrt{\dfrac{2Km}{Bq}} \Rightarrow r \propto \dfrac{\sqrt{K}}{B}$

26. Ans (4)

The energy of a charged particle moving in magnetic field remains constant because the magnetic field does not do any work. Therefore, kinetic energy is constant i.e. u = v.

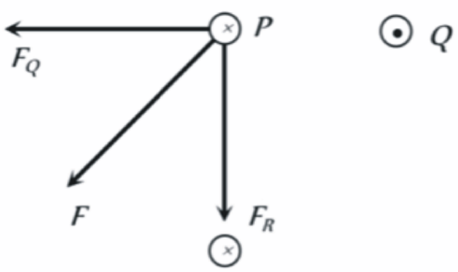
The force on electron will act along negative y-axis initially. The electron will undergo circular motion in clockwise direction and emerge out the field. So y < 0.

27. Ans (3)

$W = F.d \cos 90° = 0$

28. Ans (3)

The forces F_Q and F_R are the force applies by wires Q and R respectively on the wire P as shown in figure. Their resultant force F is best shown by C.



29. Ans (3)

By using $\vec{F}_m = q(\vec{v} \times \vec{B})$

$\Rightarrow \vec{F}_m = 2 \times 10^{-6}\{3 \times 10^6 \hat{i} \times (-0.2)\hat{k}\}$

$= 1.2(\hat{i} \times \hat{k}) = +1.2\hat{j}$

i.e.1.2 N in positive y-direction.

31. Ans (3)

$[x] = [bt^2] \Rightarrow [b] = [x/t^2] = km/s^2$

32. Ans (2)

$\tau \propto F^a v^b t^c$

$M^1L^2T^{-2} = [M^1L^1T^{-2}]^a [LT^{-1}]^b [T]^c$

$1 = a \,;\, 2 = a + b \,;\, -2 = -2a - b + c$

$\Rightarrow b = 1 \,;\, -2 = -2 + c$

$c = 1$

$\tau = FvT$

SECTION-B

37. Ans (3)

$(\mu - 1)t = \dfrac{\lambda}{2}$

$t = \dfrac{\lambda}{2(\mu - 1)}$

39. Ans (2)

Energy of γ rays = Rest mass energy of electron + Rest mass energy of positron + Kinetic energy of electron and positron

So, Energy of γ rays

$= (0.50 + 0.50 + 0.78)$ MeV

$= 1.78$ MeV

41. Ans (4)

We know that $\lambda = \dfrac{0.693}{T}$ and $-\dfrac{dN}{dt} = \lambda N$.

$\therefore$ Decay rate $= \lambda N = \dfrac{0.693}{T} \times N$

$= \dfrac{0.693 \times 4.0 \times 10^{15}}{1.2 \times 10^7}$

$\therefore$ Decay rate $= 2.3 \times 10^8$ atoms/s

43. Ans (1)

$l_{min} = \dfrac{hc}{eV} \Rightarrow \lambda \propto \dfrac{1}{V}$

$\because \lambda_2 > \lambda_1$ (see graph) $\Rightarrow V_2 > V_1$

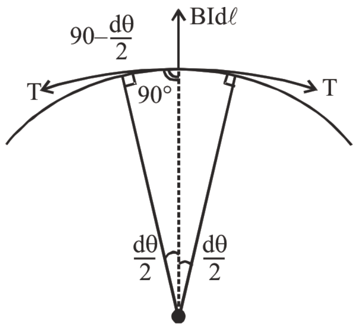
$\sqrt{v} = a(Z - b)$ (Moseley's law)

$v \;\alpha\; (z - 1)^2 \Rightarrow \lambda \propto \dfrac{1}{(Z-1)^2} \quad \left(\because v \propto \dfrac{1}{\lambda}\right)$

$\lambda_1 > \lambda_2$ (see graph for characteristic lines)

$\Rightarrow Z_2 > Z_1$

45. Ans (3)



$\frac{d\ell}{R} = d\theta$

$\Rightarrow \quad d\ell = Rd\theta$

$2T\cos\left(90° - \frac{d\theta}{2}\right) = BIdl$

$\Rightarrow \quad 2T\sin\frac{d\theta}{2} = BId\ell$

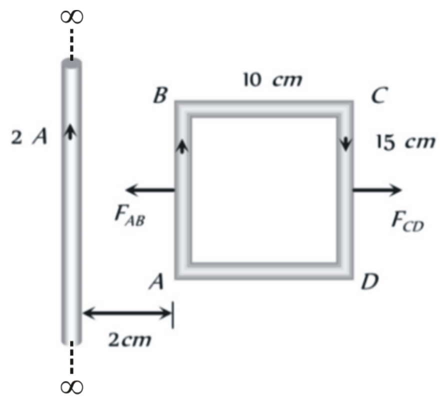
$\Rightarrow \quad 2T.\frac{d\theta}{2} = BIRd\theta$

(as $d\theta$ is very small $\sin\frac{d\theta}{2} = \frac{d\theta}{2}$)

$\Rightarrow \quad T = BIR = BI\frac{L}{2\pi} \qquad (\because L = 2\pi R)$

46. Ans (1)

Force on side BC and AD are equal but opposite so their net will be zero.



But $F_{AB} = 10^{-7} \times \frac{2 \times 2 \times 1}{2 \times 10^{-2}} \times 15 \times 10^{-2}$

$= 3 \times 10^{-6}$ N

and $F_{CD} = 10^{-7} \times \frac{2 \times 2 \times 1}{(12 \times 10^{-2})} \times 15 \times 10^{-2}$

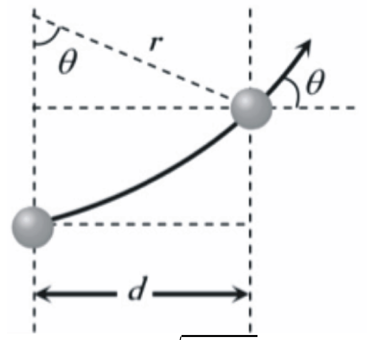
$= 0.5 \times 10^{-6}$ N

$\Rightarrow F_{net} = F_{AB} - F_{CD} = 2.5 \times 10^{-6}$ N

$= 25 \times 10^{-7}$ N, towards the wire.

47. Ans (2)

According to following figure $\sin\theta = \frac{d}{r}$



also $r = \frac{\sqrt{2mk}}{qB} = \frac{1}{B}\sqrt{\frac{2mV}{q}}$

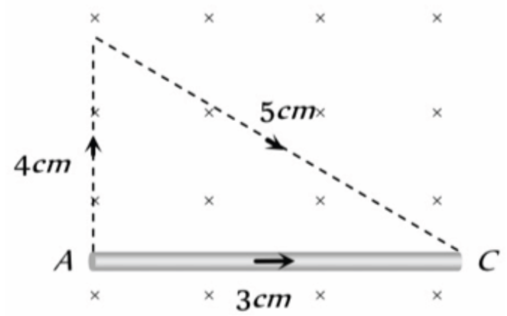
$\therefore \sin\theta = Bd\sqrt{\frac{q}{2mV}}$

$= 0.51 \times 0.1\sqrt{\frac{1.6 \times 10^{-19}}{2 \times 1.67 \times 10^{-27} \times 500 \times 10^{3}}}$

$= \frac{1}{2} \Rightarrow \theta = 30°$

48. Ans (2)

The given curved wire can be treated as a straight wire as shown



Force acting on the wire AC, F = Bil

$= 2 \times 2 \times 3 \times 10^{-2}$

$= 12 \times 10^{-2}$ N along y-axis

So acceleration of wire $= \frac{F}{m}$

$= \frac{12 \times 10^{-2}}{10 \times 10^{-3}} = 12 m/s^2$

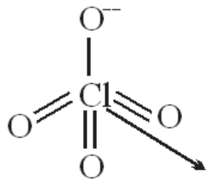
SUBJECT : CHEMISTRY

SECTION-A

51. Ans (2)

As the electronegativity of central atom increases the bond angle increases due to repulsion between bond pair and bond pair as bond pairs are more close to the central atom.

52. Ans (1)



sp^3 hybridisation of Cl.

All bonds are $p\pi\text{–}d\pi$.

53. Ans (2)

$$\underbrace{HF}_{\text{H-bonding}} < \underbrace{HI < HBr < HCl}_{\text{mol. mass }(\downarrow)\text{ volatile nature }(\uparrow)}$$

54. Ans (1)

Both N_2^+ and O_2^+ have the same bond order of 2.5.

55. Ans (2)

Concept

56. Ans (2)

$$LiNO_3 \xrightarrow{\Delta} Li_2O + NO_2 + O_2$$

57. Ans (4)

Due to small size of Be^{+2}, Polarizing power is more and their halides show covalent nature.

67. Ans (4)

NCERT-Page No. # 199

69. Ans (1)

According to question

75. Ans (4)

1 – Chain

2 – Identical

3 – Not isomer

4 – Chain (correct)

77. Ans (4)

Fact based

78. Ans (3)

NCERT-XII, Pg. # 340

79. Ans (1)

NCERT-XII, Pg. # 362

82. Ans (2)

NCERT-XII, Pg. # 382

SECTION-B

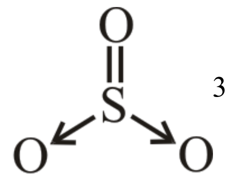
86. Ans (1)

$N \equiv N: \longrightarrow O$

B.O. = 3 maximum

$$\text{B. L.} \propto \frac{1}{\text{B. O.}}$$

87. Ans (3)



$3\sigma, 1\pi$

93. Ans (2)

$$k = \frac{1}{k_1} \times k_2$$

94. Ans (4)

$$\log \frac{k_p}{k_c} - 2 \log RT = 0$$

$$\log \frac{k_p}{k_c} = 2 \log RT$$

$$\log \frac{k_p}{k_c} = \log (RT)^2$$

$$k_p = k_c (RT)^2$$

$$\Delta n_g = 2$$

99. Ans (1)

NCERT-XII, Pg. # 363

SUBJECT : BIOLOGY-I

SECTION-A

103. Ans (1)

NCERT XII Pg.# 110

106. Ans (1)

NCERT Pg#103

107. Ans (1)

NCERT XII Page No. # 97

111. Ans (2)

NCERT Pg. # 110, Para 6.5.3

132. Ans (1)

NCERT Page no 269 & 270

SECTION-B

138. **Ans (2)**
NCERT PAGE NO. 106 (E)

140. **Ans (2)**
NCERT (XII) Pg. # 111 para 2.3

SUBJECT : BIOLOGY-II

SECTION-A

155. **Ans (2)**
NCERT XI Pg.# 298

159. **Ans (2)**
NCERT Pg # 335

165. **Ans (3)**
NCERT Pg. # 399 Para 22.4

166. **Ans (4)**
NCERT XI Pg#332

167. **Ans (3)**
NCERT Pg. # 334, First Para

169. **Ans (2)**
NCERT Pg. # 44

172. **Ans (3)**
NCERT (XIIth) Pg. # 64(E), 71,72(H)

173. **Ans (3)**
NCERT (XIIth) Pg. # 60, Para-4

174. **Ans (3)**
NCERT-XII, Pg. # 67 (H)

175. **Ans (1)**
NCERT–XII, Pg. # 62

177. **Ans (2)**
NCERT Pg.# 60, Para. 4.2

178. **Ans (3)**
NCERT (XII) Pg. # 60

179. **Ans (3)**
NCERT Pg # 47, para 3.3

180. **Ans (3)**
NCERT XII Pg # 50, 51, Para 3.4

181. **Ans (3)**
NCERT XII Pg # 64, Para 4.5

SECTION-B

187. **Ans (4)**
NCERT XIth Pg.# 332 Para-22.2.2

188. **Ans (3)**
NCERT (XI) Pg. # 334

191. **Ans (1)**
NCERT Pg. # 55

198. **Ans (4)**
NCERT XII Pg # 53, para 3.6

199. **Ans (3)**
NCERT XII Pg # 51, para 3.4